

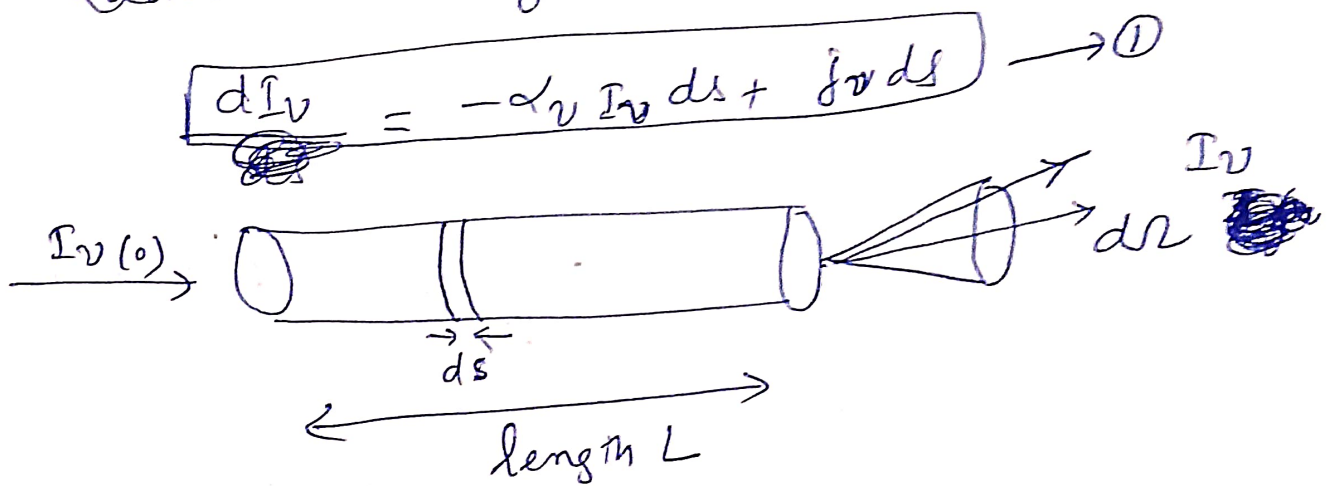
How to estimate the mass of a clond/clump/core of a molecular cloud using the dust continuum observations from single dish/interferometric telescopes!

26-27 March 2026

Reference: Kauffmann, J et al 2008
A&A, 487, 993-1017.

- Use governing radiative transfer equation
- introduce absorption coefficient (α_ν)
& specific absorption coefficient of dust (per mass) (κ_ν)
- Detailed relation b/n $\alpha_\nu, n, f, \kappa_\nu, \sigma_\nu, \rho$ and μ_m (mean molecular weight)
- Introduce the relation b/n optical depth (τ_ν), specific dust absorption coefficient (κ_ν), $\mu, m_H, N(H_2)$
- Modified radiative transfer eqn considering absence of background emission and optically thin case
- Relation b/n specific intensity, I_ν , and flux, S_ν^{beam} .
- Solid angle of the beam Ω_{beam}
- Write the $B_\nu(T)$ in terms of useful parameters
- $I_\nu = B_\nu(T) \tau_\nu$, $\tau_\nu = \kappa_\nu \mu m_H N(H_2)$, $B_\nu(T) = \frac{2h\nu^3}{12} \frac{1}{\left[\exp\left(\frac{h\nu}{kT}\right) - 1 \right]}$
- $S_\nu^{beam} = I_\nu \Omega_{beam}$, $\Omega_{beam} \approx 1.133 \theta_{HPBW}^2$
- Get Relation b/n S_ν^{beam} , $N(H_2)$, κ_ν , θ_{HPBW} , T , in useful parameters.

① Radiative transfer equation:



$$\frac{dI_v}{\alpha_v ds} + I_v - \frac{j_v}{\alpha_v} = 0 \quad \left| \quad \tau_v = \alpha_v ds = \frac{1}{\text{cm}} \cdot \text{cm} = \text{no units} \right.$$

$$\frac{dI_v}{d\tau_v} + I_v = \frac{j_v}{\alpha_v} = S_v = \text{Source term}$$

$$\boxed{\frac{dI_v}{d\tau_v} + I_v = S_v}$$

First-order linear differential equation

• Solution? $I_v = ?$

Let us introduce the integrating factor

$$M(\tau_v) = e^{\int d\tau_v} = e^{\tau_v}$$

$$e^{\tau_v} \frac{dI_v}{d\tau_v} + e^{\tau_v} I_v = e^{\tau_v} S_v$$

$$\frac{d}{d\tau_v} (I_v e^{\tau_v}) = S_v e^{\tau_v}$$

$$\int_0^{\tau_v} \frac{d}{d\tau_v} (I_v(\tau') e^{\tau'}) d\tau' = \int_0^{\tau_v} S_v(\tau') e^{\tau'} d\tau'$$

$$I_v(\tau') e^{\tau'} - I_v(0) = \int_0^{\tau_v} S_v(\tau') e^{\tau'} d\tau'$$

$$= \int_0^{\tau_v} S_v(\tau') e^{\tau'} d\tau' = \int_0^{\tau_v} S_v(\tau') e^{\tau'} d\tau'$$

$$I_v(\tau_v) e^{\tau_v} - I_v(0) = \underline{\hspace{2cm}}$$

Ex

If $M(x) = e^{x^2}$ then

$$\frac{dM}{dx} = e^{x^2} \cdot 2x$$

$$\frac{dM}{dx} = 2x e^{x^2} = 2x M(x)$$

If $P(x) = 2x$, then

$$\frac{dM}{dx} = \cancel{2x} P(x) M(x)$$

such that

$M(x) = e^{x^2}$ is valid for only eqns of the form

$$\frac{dy}{dx} + 2xy = Q(x)$$

$$\boxed{\frac{dy}{dx} + P(x)y = Q(x)}$$

integrating factor

$$M(x) = e^{\int P(x) dx}$$

$$I_\nu(\tau_\nu) = I_\nu(0) e^{-\tau_\nu} + e^{-\tau_\nu} \int_0^{\tau_\nu} S_\nu(\tau') e^{\tau'} d\tau'$$

Background
emission set
attenuated by the
intervening medium
with optical depth τ_ν

source emission
+ self absorption

$$I_\nu(\tau_\nu) = I_\nu(0) e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(\tau') e^{-(\tau_\nu - \tau')} d\tau'$$

If the medium is kept at local thermodynamical equilibrium (LTE)

$S_\nu = B_\nu(T)$, the medium is isothermal with a constant T

$$I_\nu = I_\nu(0) e^{-\tau_\nu} + B_\nu(T) \underbrace{\int_0^{\tau_\nu} e^{-(\tau_\nu - \tau')} d\tau'}_{1 - e^{-\tau_\nu}}$$

~~$I_\nu = B_\nu(T)$~~

$$I_\nu = I_\nu(0) e^{-\tau_\nu} + B_\nu(T) (1 - e^{-\tau_\nu}) \rightarrow \textcircled{1}$$

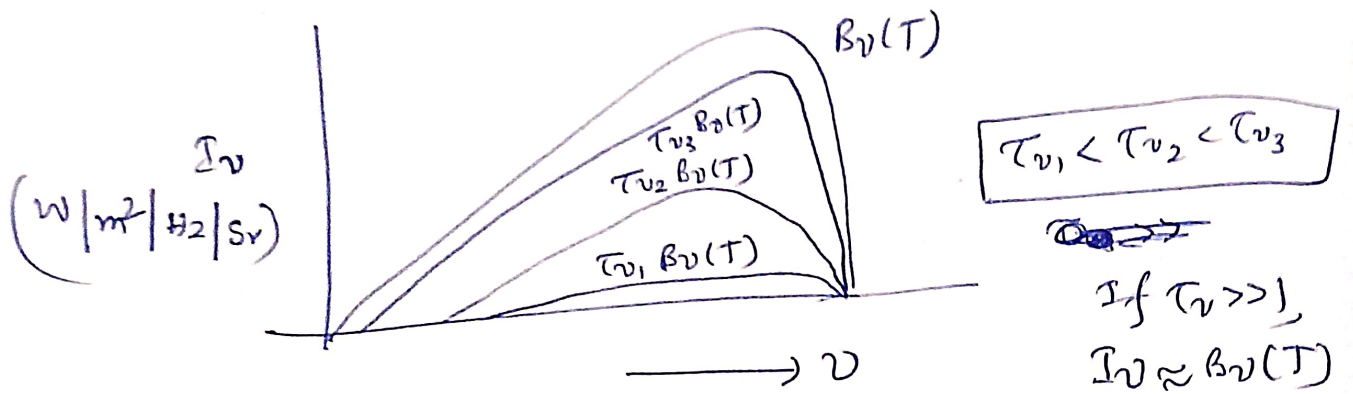
If there is no background emission: $I_\nu(0) e^{-\tau_\nu} = 0$

$$I_\nu = B_\nu(T) (1 - e^{-\tau_\nu})$$

for optically thin emission $\tau_\nu \ll 1$, $e^{-\tau_\nu} = 1 - \tau_\nu + \frac{\tau_\nu^2}{2} - \frac{\tau_\nu^3}{6} + \dots$
 $e^{-\tau_\nu} \approx 1 - \tau_\nu$

$$I_\nu \approx \tau_\nu B_\nu(T)$$

$\rightarrow \textcircled{2}$



* optical depth (T_v) its relation with
 $\alpha_v, \sigma_v, n, \rho, K_v, \mu_m$, etc

Absorption
Coefficient

$$\boxed{\alpha_v = n \sigma_v} \rightarrow (3)$$

number density = $\frac{\text{no. of particles}}{\text{unit volume}} = \text{cm}^{-3}$

σ_v = cross sectional area of
each particle
= cm^2

$$\boxed{\alpha_v = \text{cm}^{-3} \cdot \text{cm}^2 = \text{cm}^{-1}}$$

$$\frac{dI_v}{I_v} = -\alpha_v ds$$

$$\boxed{dI_v = -\alpha_v I_v ds}$$

$$\text{mass density } \rho = n \mu m_H$$

$$n = \frac{\rho}{\mu m_H} \rightarrow (4)$$

μ = mean molecular
weight

m_H = mass of hydrogen
atom

(3) & (4)

$$\alpha_v = \frac{\rho}{\mu m_H} \sigma_v = \left(\frac{\sigma_v}{\mu m_H} \right) \rho$$

$$\boxed{K_v = \frac{\sigma_v}{\mu m_H} = \frac{\text{cm}^2}{g}}$$

$$\boxed{\alpha_v = K_v \rho} \rightarrow (5)$$

where K_v = specific absorption coefficient
(absorption coefficient/unit mass)

Let us recall

$$dI_\nu = -\alpha_\nu I_\nu ds$$

$$\frac{dI_\nu}{I_\nu} = -\alpha ds$$

$$\frac{dI_\nu}{I_\nu} = -d\tau_\nu$$

$$d\tau_\nu \equiv \alpha_\nu ds \rightarrow \textcircled{5}$$

④ & ⑤

$$d\tau_\nu = K_\nu \rho ds$$

$$\int d\tau_\nu = \int K_\nu \rho ds$$

$$\tau_\nu = \int K_\nu \rho ds = K_\nu \mu m_H N(H_2) \quad (*)$$

$$\tau_\nu = \int K_\nu n_{H_2} \mu m_H ds = K_\nu \mu m_H \int n_{H_2} ds$$

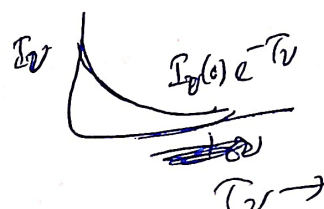
$$\tau_\nu = K_\nu \mu m_H N(H_2) \quad \textcircled{6}$$

$$\frac{dI_\nu}{I_\nu} = -d\tau_\nu$$

$$\int \frac{dI_\nu}{I_\nu} = -\int d\tau_\nu$$

$$\ln I_\nu = -\tau_\nu + C$$

$$I_\nu = I_{\nu(0)} e^{-\tau_\nu}$$



from ② & ⑥ : $I_\nu \equiv \tau_\nu B_\nu(T)$; $\tau_\nu = \int K_\nu \rho ds$

$$I_\nu = \left(\int K_\nu \rho ds \right) B_\nu(T)$$

Planck
function

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1} \quad W/m^2/Hz/Sr$$

ν in terms of GHz :

$$\nu = \left(\frac{\nu}{GHz} \right) \times 10^9$$

$$\nu^3 = \left(\frac{\nu}{GHz} \right)^3 \times 10^{27}$$

$$B_\nu(T) = \left(\frac{2h}{c^2} \right) \times \left(\frac{\nu}{GHz} \right)^3 \times 10^{27} \cdot \frac{1}{e^{h\nu/kT} - 1}$$

$$\frac{2h}{c^2} = \frac{2 \times 6.626 \times 10^{-34} \text{ J}\cdot\text{s}}{(3 \times 10^8 \text{ m/s})^2} = 1.47 \times 10^{-50}$$

$$c = 2.99 \times 10^8 \text{ m/s}$$

$$\left(\frac{2h}{c^2} \right) \times 10^{27} = 1.475 \times 10^{-23}$$

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1}$$

$$\begin{aligned} \frac{2h\nu^3}{c^2} &= 2h\left(\frac{c}{\lambda}\right)^3 \times \frac{1}{c^2} = \frac{2hc}{\lambda^3} = 2 \times 6.626 \times 10^{-34} \times 3 \times 10^8 \times \left(\frac{1}{\lambda}\right)^3 \\ &= " " " \times \left(\frac{1 \text{ mm}}{\lambda \times 10^{-3}}\right)^3 \\ &= " " " \left(\frac{1 \text{ mm}}{\lambda}\right)^3 \times 10^9 \\ &= 2 \times 6.626 \times 10^{-34} \times 3 \times 10^8 \times \left(\frac{\lambda}{1 \text{ mm}}\right)^{-3} \times 10^9 \end{aligned}$$

$$\boxed{\frac{2h\nu^3}{c^2} = 3.975 \times 10^{16} \times \left(\frac{\lambda}{1 \text{ mm}}\right)^{-3}}$$

$$\frac{h\nu}{k_B T} = \frac{hc}{\lambda k_B T} = \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{1.381 \times 10^{-23}} \left(\frac{1}{\lambda}\right) \times \left(\frac{1}{T}\right)$$

$$\begin{aligned} \left(\frac{1}{\lambda} = \left(\frac{1 \text{ mm}}{\lambda \times 10^{-3}}\right) = \left(\frac{\lambda}{1 \text{ mm}}\right)^{-1} \times 10^3 \right) & \left(\left(\frac{10 \text{ K}}{T} \times \frac{1}{10}\right) = \left(\frac{10 \text{ K}}{T}\right) \times \frac{1}{10} \right. \\ & \left. = \left(\frac{T}{10 \text{ K}}\right)^{-1} \times \frac{1}{10} \right) \\ & = \left(\frac{6.625 \times 10^{-34} \times 3 \times 10^8 \times 10^2}{1.381 \times 10^{-23}} \right) \times \left(\frac{\lambda}{1 \text{ mm}}\right)^{-1} \times \left(\frac{T}{10 \text{ K}}\right)^{-1} \end{aligned}$$

$$\boxed{\frac{h\nu}{k_B T} = 1.43917 \times \left(\frac{\lambda}{1 \text{ mm}}\right)^{-1} \times \left(\frac{T}{10 \text{ K}}\right)^{-1}}$$

$$\boxed{B_\nu(T) = 3.975 \times 10^{16} \times \left(\frac{\lambda}{1 \text{ mm}}\right)^{-3} \times \frac{1}{\left[e^{1.43917 \times \left(\frac{\lambda}{1 \text{ mm}}\right)^{-1} \times \left(\frac{T}{10 \text{ K}}\right)^{-1}} - 1 \right]}}$$

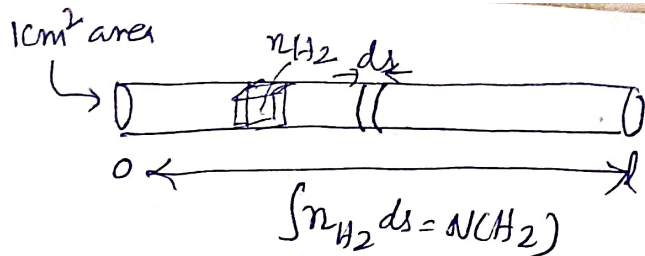
→ ⑦

$$T_\nu = K_\nu \int \rho ds$$

$$= K_\nu \int n_{H_2} \mu_{H_2} m_H ds$$

$$= K_\nu \mu_{H_2} m_H \int n_{H_2} ds$$

$$= K_\nu \mu_{H_2} m_H N(H_2)$$



$$n_{H_2} = \text{number density} = /cm^3$$

$$N(H_2) = \text{column density} = /cm^2$$

$$= \left(\frac{K_\nu}{0.01 \frac{cm^2}{g}} \right) \times 0.01 \times 2.8 \times 1.67 \times 10^{-24} g \times \left(\frac{N(H_2)}{2.02 \times 10^{20}} \right) \times \left(\frac{2.02 \times 10^{20}}{10^{20}} \right)$$

$$T_\nu = (9.44552 \times 10^{-6}) \left(\frac{K_\nu}{0.01 \frac{cm^2}{g}} \right) \times \left(\frac{N(H_2)}{2.02 \times 10^{20}} \right)$$

$$F_\nu^{beam} = T_\nu \Omega_{beam}$$

beam solid angle (Sr)

specific intensity ($W/m^2/Hz/Sr$)



$$F_\nu^{beam} = B_\nu(T) T_\nu \Omega_B$$

$$F_\nu^{beam} = T_\nu B_\nu(T) \Omega_B$$

eg ⑧, ⑨, & ⑩ into ⑩

$$F_\nu^{beam} = 3.975 \times 10^{-16} \times \left(\frac{\lambda}{1mm} \right)^{-3} \times$$

$$\left\{ \frac{1}{2^{1.4398} \times \left(\frac{\lambda}{1mm} \right)^{-1} \times \left(\frac{T}{10K} \right)^{-1} - 1} \right\}$$

$$\times 2.665 \times 10^9 \times \left(\frac{\theta_{HPBW}}{10''} \right)^2 \times$$

$$\Omega_B = 1.33 \times \theta_{HPBW}^2$$

$$= 2.665 \times 10^9 Sr \times \left(\frac{\theta_{HPBW}}{10 \text{ arcsec}} \right)^2$$

HPBW = Half Power Beam Width

$$\Omega_B = 2.665 \times 10^9 Sr \times \left(\frac{\theta_{HPBW}}{10 \text{ arcsec}} \right)^2$$

$$\left(9.44552 \times 10^{-6} \right) \times \left(\frac{N(H_2) cm^2}{2.02 \times 10^{20}} \right) \times \left(\frac{K_\nu}{0.01} \right)$$

$$F_{\nu}^{\text{beam}} = \frac{1.0006 \times 10^{-29}}{\text{g}} \times \left(\frac{1 \text{ mm}}{\lambda}\right)^3 \times \left[e^{1.439 \times \left(\frac{1 \text{ mm}}{\lambda}\right) \times \left(\frac{10 \text{ K}}{T}\right)} - 1 \right]^{-1} \times \left(\frac{K_{\nu}}{0.01 \frac{\text{cm}^2}{\text{g}}}\right) \times \left(\frac{\theta_{\text{HPBW}}}{10 \text{ arcsec}}\right)^2 \times \left(\frac{N(\text{H}_2)}{2.02 \times 10^{20} \text{ cm}^{-2}}\right)$$

$$\frac{1 \text{ mJy}}{\text{beam}} = 1 \times 10^{-3} \times 10^{-26} \times \text{W/m}^2 / \text{H}_2 / \text{beam}$$

$$\left(\frac{F_{\nu}^{\text{beam}}}{1 \text{ mJy/beam}} \right) = \left(\frac{N(\text{H}_2)}{2.02 \times 10^{20} \text{ cm}^{-2}} \right) \times \left(\frac{K_{\nu}}{0.01 \frac{\text{cm}^2}{\text{g}}} \right) \times \left(\frac{\theta_{\text{HPBW}}}{10''} \right)^2 \times \left(\frac{1 \text{ mm}}{\lambda} \right)^3 \times \left[e^{1.43912 \times \left(\frac{1 \text{ mm}}{\lambda}\right) \times \left(\frac{10 \text{ K}}{T}\right)} - 1 \right]^{-1}$$

$$N(\text{H}_2) = \left(2.02 \times 10^{20} \text{ cm}^{-2} \right) \times \left(e^{1.439 \left(\frac{\lambda}{\text{mm}}\right)^{-1} \left(\frac{T}{10 \text{ K}}\right)^{-1}} - 1 \right)^{-1} \times \left(\frac{\lambda}{\text{mm}} \right)^{-3} \times \left(\frac{K_{\nu}}{0.01 \frac{\text{cm}^2}{\text{g}}} \right)^{-1} \times \left(\frac{F_{\nu}^{\text{beam}}}{\text{mJy/beam}} \right) \times \left(\frac{\theta_{\text{HPBW}}}{10''} \right)^{-2}$$

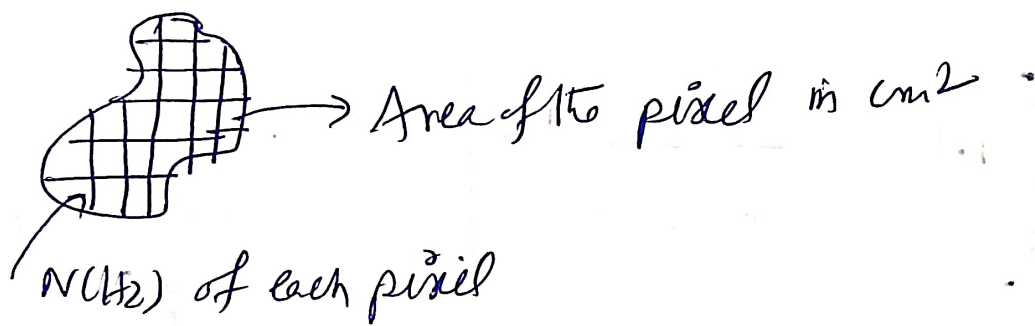
at every frequency we could estimate K_{ν}
 we have F_{ν}^{beam} (flux/beam = mJy/beam),
 we can estimate/derive T & $N(\text{H}_2)$ using
 non-linear least square fitting method

How to estimate Mass

Mass is given by the integral of the $N(\text{H}_2)$ across the source

$$M = \mu_{\text{H}_2} m_{\text{H}} \int N(\text{H}_2) dA \quad \longrightarrow \textcircled{1}$$

(column density of a pixel. $\times$ area of the pixel)



$\int N(\text{H}_2) dA = \text{total integrated column density}$

We know: $\tau_\nu = \mu_{\text{H}_2} m_{\text{H}} K_\nu N(\text{H}_2)$ or $N(\text{H}_2) = \frac{\tau_\nu}{\mu_{\text{H}_2} m_{\text{H}} K_\nu}$

$$I_\nu \cong B_\nu(T) \tau_\nu \quad \longrightarrow \textcircled{2}$$

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{\left[\exp(h\nu/kT) - 1 \right]} \quad \longrightarrow \textcircled{4}$$

① & ②:

$$M = \mu_{\text{H}_2} m_{\text{H}} \int \frac{\tau_\nu}{\mu_{\text{H}_2} m_{\text{H}} K_\nu} dA$$
$$= \int \frac{\tau_\nu dA}{K_\nu} = \int \frac{I_\nu}{B_\nu(T)} \cdot \frac{1}{K_\nu} dA$$
$$\boxed{M = \frac{1}{K_\nu B_\nu(T)} \int I_\nu dA} \quad \longrightarrow \textcircled{5}$$

$$M = \frac{1}{K_\nu B_\nu(T)} \int I_\nu dA$$

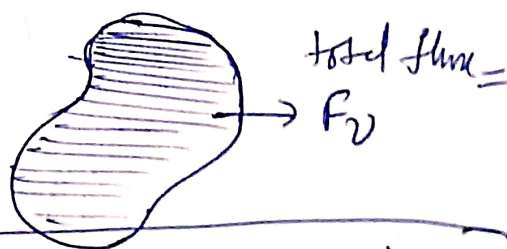
dA = pixel area of the surface element dA is related to the solid angle element $d\Omega$ by

$$dA = d^2 d\Omega ; \quad d = \text{distance to the source}$$

$$M = \frac{d^2}{K_\nu B_\nu(T)} \int I_\nu d\Omega$$

$$M = \frac{d^2}{K_\nu B_\nu(T)} F_\nu ; \quad \text{where } F_\nu = \int I_\nu d\Omega$$

is the integrated flux



$$M = 0.12 M_\odot \left[e^{1.439 \left(\frac{\lambda}{\text{mm}} \right)^{-1} \left(\frac{T}{10\text{K}} \right)^{-1}} - 1 \right] \times \left(\frac{K_\nu}{0.01 \text{ cm}^2/\text{g}} \right)^{-1} \times$$

$$\left(\frac{F_\nu}{\text{Jy}} \right) \left(\frac{d}{100 \text{ pc}} \right)^2 \left(\frac{\lambda}{\text{mm}} \right)^3$$